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Case Report

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Assessing the Effectiveness of Virtual Reality Serious Games in Post-Stroke Rehabilitation: A Novel Evaluation Method

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Abstract

Background: This paper presents a novel evaluation method for assessing the effectiveness of virtual reality serious games in post-stroke rehabilitation. Stroke, which is a leading cause of long-term disability, requires effective rehabilitation approaches to facilitate motor recovery and enhance patients' quality of life. The study aims to investigate the potential benefits of incorporating VR technology into upper limb rehabilitation for post-stroke patients, with a specific focus on evaluating patient motivation and engagement. The proposed evaluation method provides insights into the effectiveness of VR serious games as a rehabilitation tool and contributes to the understanding of their impact on post-stroke recovery.
Methods: This study involved the recruitment of a diverse sample of 20 patients,

both male and female, who had experienced left hemiplegia as a result of an ischemic stroke. To ensure a representative sample, patients of different ages were included. Participants were divided into three groups, with each group invited to participate in different stages over a one-year period. During an eight-week period, all groups participated in a rehabilitation program consisting of virtual reality exercises twice a week, alongside five traditional rehabilitation sessions per week. Furthermore, this study introduces a novel approach using camera-based sensors to evaluate patient engagement in virtual reality exercises. By monitoring and analyzing facial expressions throughout each rehabilitation session, our objective is to leverage emotion recognition technology to understand patients' emotional states and levels of motivation and engagement.
Results: The findings from this study indicate that a significant number of patients showed motivation during virtual reality rehabilitation exercises.
Conclusion: The analysis of facial expressions utilizing camera-based sensors,yielded valuable insights into the emotional state and level of engagement exhibited by the participants.These findings substantiate the efficacy of virtual reality as a potent tool in fostering motivation among post-stroke patients, thereby actively involving them in rehabilitation exercises and facilitating their journey toward recovery.

Keywords: Emotion Recognition, Facial Emotion, Motivation, Post-Stroke, Rehabilitation, virtual reality, serious games, Evaluation Method

1 Introduction

Strokes, also known as cerebrovascular accidents (CVA), are medical emergencies that occur when the blood flow to the brain is interrupted or reduced. This can happen due to a blocked blood vessel (ischemic stroke) or a ruptured blood vessel (hemorrhagic stroke) [1]. In Algeria, approximately 60,000 new cases of stroke are recorded annually, but only one in every 200 patients receives emergency treatment.The absence of proper care leads to an approximate annual death toll of 20,000. [2]. Patients who have suffered a stroke typically experience a range of motor, cognitive, and sensory impairments. According to [3], more than 60% of stroke survivors are estimated to endure long-term side effects throughout their lives. These effects encompass conditions such as paralysis, memory loss, and speech difficulties.Therefore, rehabilitation plays a critical role in facilitating motor recovery in stroke patients.

Stroke rehabilitation often takes a long time, sometimes in some cases lasting a lifetime. Positive emotions during therapy and patient motivation are crucial for successful rehabilitation and facilitating recovery after a stroke, emphasizing that positive treatment outcomes depend not only on the efforts of healthcare professionals but also on the willingness, active participation, and motivation of the patient. In contrast, Research has shown that positive emotions lead to positive therapeutic outcomes, while negative emotions can result in unfavorable outcomes [4–7]; Low motivation in stroke patients has been shown to increase the difficulty of rehabilitation [8, 9]. To

assist this large number of stroke victims, new therapy methods such as virtual reality (VR) and interactive video games (IVG) have been developed [10–12]. Utilizing computer simulations of real-world items and scenarios is a common therapy technique to optimize brain plasticity mechanisms and promote movement recovery and other functions [13]. VR therapy has been shown to offer significant benefits to stroke patients, particularly by incorporating "serious games" into the rehabilitation process. Such games can increase patient motivation to participate in and repeat exercises, as well as foster greater engagement with therapeutic activities recommended by medical professionals [14, 15] . In the present study, we aim to investigate stroke patients' motivations for incorporating VR into their rehabilitation process. Specifically, we seek to examine whether the use of VR technology can lead to greater patient stimulation and faster recovery compared to conventional rehabilitation methods. Our research will therefore help to test this hypothesis and contribute to a better understanding of the potential benefits of VR in stroke rehabilitation.

2 Related Works

It is also generally agreed that VR or serious games can increase a person's motivation to practice more frequently and/or intensively because engaging in virtual scenarios provides enthusiasm, challenge, and pleasure [13, 15, 16]. For this purpose, several studies were carried out about the effectiveness of virtual rehabilitation applications as well as to explore how motivation, pleasure, and commitment were described or measured in VR studies in post stroke rehabilitation. In recent years, there have been several studies on the use of the Leap Motion Controller system in VR environments to improve motor function in stroke patients' upper extremities [10, 17–19], where researchers have created VR protocols based on daily activities and traditional rehabilitation exercises to create a more engaging and functional rehabilitation experience for patients.

Studies on the use of VR in stroke rehabilitation have been conducted across different phases of the condition, including the acute [20, 21] and sub-acute [19, 22] phases, as well as in chronic stroke patients [11, 12]. These studies aimed to investigate the effectiveness of VR in improving upper extremity motor function in stroke survivors, as well as to assess its feasibility, motivation, and satisfaction. Patients who used the virtual rehabilitation system reported increased motivation compared to those who received conventional rehabilitation, and high levels of satisfaction and attendance were observed in the VR groups. Researchers relied on questionnaires to assess satisfaction and attendance rates in their work. Although the results of these studies are promising, they have many restrictions to mention. Firstly, the sample sizes are small; most of these studies tested their virtual rehabilitation platforms on a few number of patients (larger random clinical trials are required to validate these preliminary results). Secondly, the sample is not homogeneous (Not all patients suffer the same type of stroke) this may distort the reported results. Finally, patients with cognitive impairments have not been thoroughly assessed, which may affect their ability to engage in VR therapy.

Other studies [23–25] looked at how the RV affected stroke patients' motivation

and cognitive function. They randomly selected 12 subjects [23]; in another experimental study, 20 patients [24], where they used VR rehabilitation protocols and compared them to a group that underwent conventional rehabilitation exercises; the VR group was more effective in improving motor impairment and attention, leading the researchers to conclude that combining conventional and virtual rehabilitation is necessary for improving patients' functional outcomes.

Several researchers have demonstrated in recent clinical trials with stroke survivors that VR is effective for the rehabilitation of motor function [15, 26–28]; performance and independence in activities of daily life [29, 30] and dynamic balance [4–6]. Also, in one of the largest studies conducted so far, which included 376 patients, it was shown that VR-based rehabilitation was beneficial [31]. However, a systematic review based on the analysis of 22 research, involving 1038 stroke patients, showed that virtual rehabilitation was not more effective than conventional therapy for enhancing upper-limb function [32]. Another review analyzed 33 studies involving 971 participants, 492 of whom used VR; and found small to moderate effects compared to people who underwent conventional therapy [33]. It is important to note that these results are not conclusive, and there is no consensus on matter, as the results are not statistically significant. In contrast, another systematic review focused on the effect of motivation on the outcomes of virtual rehabilitation programs in children and adolescents with cerebral palsy [34]. The study examined 178 articles involving 95 patients, and the results concluded that there is a paucity of research on the effects of motivation in patients using VR for rehabilitation. This finding is also supported by other recent studies [35–38], which highlighted that the use of VR does not improve the performance of stroke patients. Given the conflicting and inconclusive results of previous studies, some researchers have suggested using different approaches to evaluate the effect of VR on stroke patients. An experimental study by Zun-rong Wang et al. included 26 subacute stroke patients, who were divided into two groups: the experimental group, which received VR therapy in addition to conventional rehabilitation, and the control group, which only received conventional rehabilitation [39]. Functional magnetic resonance imaging (FMRI)[40] was used to detect brain activation, and the motor function of the affected upper limb was evaluated using the Wolf motor function test (WMFT) [41]. May be one of the benefits of using VR in rehabilitation, is the activation of different brain areas, including the mirror neuron system responsible for motor planning and learning, as shown by studies using electroencephalographic (EEG) oscillation analysis [28, 42, 43]. However, a study that also utilized EEG analysis found no significant difference in motivation or participation between conventional and VR rehabilitation approaches [44].

The conflicting and inconclusive results on the effectiveness of VR to motivate and aid stroke patients in motor rehabilitation can be attributed to the lack of homogeneity of approaches and tools used in VR studies to evaluate its effectiveness. This paper included relevant studies, and their results were summarized in table 1. Furthermore, as motivation is a multifaceted construct, it is challenging to find a way to measure it or protocols to determine its effectiveness in rehabilitation. Thus, the critical question that remains is whether VR interventions increase patient motivation and their effectiveness in motor rehabilitation.

This paper introduces a novel method for objectively assessing motivation levels in post-stroke patients using VR technology and emotion recognition. It also develops a dedicated VR-based rehabilitation platform to facilitate the practical implementation of the assessment method and enhance the understanding of motivation in post-stroke rehabilitation. Accordingly, the main contributions of this paper can be summarized as follows:

- Introduced a novel method for objectively assessing motivation levels in post-stroke patients.
- Leveraged the capabilities of emotion recognition technology to delve deeper into the emotional responses and engagement of patients during VR-based rehabilitation exercises.
- Provide a more comprehensive understanding of how VR impacts patient motivation in the context of post-stroke rehabilitation.
- Developed a dedicated VR-based rehabilitation platform as an integrated framework for delivering VR exercises and capturing emotional data simultaneously.

The remaining sections of the paper are structured as follows. After providing an overview of relevant works on serious game rehabilitation in Sections 1 and 2, section 3 outlines the proposed method, including the recruitment process of participants and the evaluation protocol. Section 4 focuses on the system design, highlighting the key components and features of the proposed system. The virtual environmental scenes used in the rehabilitation process are described in Section 5. In Section 6, we present the results obtained from the study and engage in a detailed discussion to analyze and interpret the patients' behavior during the rehabilitation process. Finally, Section 7 concludes the paper by summarizing the main findings and highlighting the effects of VR in rehabilitation.

Table 1: Summary of Studies on the Use of VR in Stroke Rehabilitation: Impact on Patient Motivation

Study	Sample	Stroke type	Evaluation protocols	Outcome
[15]	50 patients (25 underwent VR training and 25 underwent conventional therapy)	Random sample	- Montreal Cognitive Assessment (MoCA); - Functional Independence Measure (FIM); - Fugl Mayer Assessment Upper Extremity Scale (FMA-UE).	VR is more effective than conventional therapy in promoting upper-limb function recovery.
[9]	54 patients (22 underwent VR training and 32 underwent conventional therapy)	Random sample	-Box and Block Test; Barthel; Line Bisection Test (LBS).	No significant differences indicating that VR is better than the conventional therapy.
[16]	A systematic review based on an analysis of 54 papers.	Random sample		The results of the use of VR were inconclusive.
[13]	24 patients	Random sample	Fugl-Meyer Upper Extremity (FMUE). manual muscle testing.	VR is more effective than conventional therapy in promoting upper-limb function recovery.
[17]	10 patients	09 ischemic Strokes and 01 hemorrhagic stroke.	Action Research Arm Test (ARAT). Block and Box Test (BBT). satisfaction questionnaire and attendance rate.	VR is an effective tool in rehabilitation and gives high levels of attendance and satisfaction to patients.
[18]	10 patients (5 underwent VR training and 5 underwent conventional therapy)	Random sample	Motivation questionnaire; Box and Block.	Compared to the control group, the virtual rehabilitation group showed somewhat higher motivation levels.
[19]	4 elderly patients	Ischemic stroke	Pittsburgh Scale	VR has shown to be a source of motivation and has played a significant role in enhancing the rehabilitation progress of hand abilities.
[20]	13 stroke patients	Random sample	Questionnaire	Overall, VR therapy left a positive impression.
[21]	15 healthy individuals and 15 stroke patients	Random sample	Questionnaire	VR is motivating
[22]	11 subject	Random sample	-Questionnaire; fugl-meyer scale.	VR is motivating
[11]	65 patients (33 underwent VR training and 32 underwent conventional therapy)	Random sample	-Functional Independence Measure (FIM); Fugl-Meyer Upper Extremity Scale (FMUE); Performance Assessment of Self-Care Skills (PASS).	VR rehabilitation is effective in improving upper extremity function and self-care skills, but it did not improve functional independence.
[23]	12 subjects	Random sample	clinical-psychometric battery scale	VR can be considered a beneficial complementary treatment.

[24]	20 patients (10 underwent VR training and 10 underwent conventional therapy)	Random sample	- Cognitive Assessment Test (MoCA); Functional Independence Measure (FIM); Motricity Index (MI).	VR is motivating
[26]	30 patients (15 underwent VR training and 15 underwent conventional therapy)	Random sample	FuglMeyer scale	Virtual rehabilitation effectively treat post-stroke hemiparetic patients
[35]	40 patients (20 underwent VR training and 20 underwent conventional therapy)	Random sample	FuglMeyer scale	No significant differences indicating that VR is better than the conventional therapy.
[36]	20 patients (10 underwent VR training and 10 underwent conventional therapy)	Random sample	FuglMeyer scale	No significant differences indicating that VR is better than the conventional therapy
[27]	5 patients	Hemiplegics stroke	FuglMeyer scale	Improved performance after VR therapy.
[28]	18 patients	Ischemic stroke	neuroimaging	Improved performance after VR therapy.
[29]	10 patients	Random sample	- Barthel scale; FuglMeyer scale.	Improved performance after VR therapy.
[30]	50 patients (25 underwent VR with conventional therapy and 25 underwent conventional therapy only)	Random sample	- Berg Balance Scale-BBS; BARTHEL INDEX-BI.	VR in addition to conventional therapy is more effective than conventional therapy alone.
[5]	10 patients (5 underwent VR with conventional therapy and 5 underwent conventional therapy only)	Random sample	Using the System Usability Scale (SUS).	VR in addition to conventional therapy is more effective than conventional therapy alone.
[6]	50 patients (25 underwent VR with conventional therapy and 25 underwent conventional therapy only)	Random sample	-Berg Balance Scale (BBS); Functional Reach Test; Barthel Index.	The VR group exhibited a greater level of pleasure compared to the conventional group.
[31]	376 patients (263 underwent VR with conventional therapy and 133 underwent conventional therapy only)	Random sample	- Fugl-Meyer Upper Extremity (F-M UE); Functional Independence Measure (FIM).	VR rehabilitation has shown superior effectiveness in restoring upper limb motor impairments compared to conventional interventions.

[23]	12 subjects	Random sample	clinical-psychometric battery scale	VR can be considered a beneficial complementary treatment.
[32]	A systematic review based on an analysis of 22 research, encompassing 1038 stroke patients.	Random sample	-	The effectivenfess of VR therapy in enhancing upper-limb function was not superior to conventional therapy.
[33]	A systematic review based on an analysis of 33 studies with 971 participants (492 underwent VR with conventional therapy)	Random sample	-	small to medium effects have been seen on VR users.
[34]	A systematic review based on an analysis of 178 studies with 95 participants.	Random sample	-	A paucity of research on the effects of motivation in patients who used VR to rehabilitate.
[39]	26 patients	Random sample	Wolf Motor Function Test (WMFT).	VR training demonstrated promising outcomes in facilitating the recovery of motor functions.
[42]	24 patients	ischemic stroke	Electroencephalographic (EEG) oscillations.	VR improves motor performance.
[43]	08 patients	Random sample	Electroencephalographic (EEG) oscillations.	VR improves motor performance.
[44]	40 patients	Random sample	Electroencephalographic (EEG) oscillations.	No significant differences between conventional and VR rehabilitation.
[37]	141 patients	Random sample	Wolf Motor Function Test (WMFT).	No significant differences indicating that VR is better than the conventional therapy
[38]	A systematic review based on an analysis of 27 papers.	Random sample	-	No significant differences indicating that VR is better than the conventional therapy

3 Methods

3.1 Recruitment of participants

Participant recruitment is a critical step in any research study, and it is particularly important in studies investigating the effectiveness of VR for motor recovery post-stroke. In this study, we recruited 20 patients who had suffered from an ischemic stroke resulting in left hemiplegia. Patients were of different sexes and ages to ensure a representative sample. We divided the patients into three groups, with each group invited to participate in the study at different stages over the course of a full year. Each group participated in rehabilitation exercises for eight weeks, and during this period, they performed VR exercises twice a week in addition to five traditional rehabilitation sessions per week. This intervention period was chosen on the basis of previous research suggesting that an eight-week intervention period is sufficient to observe measurable changes in motor function and that longer periods may not provide additional benefits [25, 32, 37, 38]. Inclusion criteria for participant selection in this study were that patients must have experienced a stroke at least 6 months prior to registration and must have exhibited motor deficits in their upper limbs. The participants were provided with information regarding the study design and their rights. We also excluded patients with significant cognitive impairment or medical comorbidities that would limit their ability to participate in the study.

3.2 Evaluation Protocol

Table 1 provides an overview of previous studies investigating the effectiveness of VR for motor recovery post-stroke. Each study includes information on the number of patients, the type of stroke, the methods used to measure patient motivation, and the main outcome of the study. The majority of these studies rely on self-report questionnaires to assess patient motivation. Measuring patient motivation in serious rehabilitation games can be challenging, as traditional self-report questionnaires may not accurately capture patient motivation's complex and dynamic nature. In addition, patients may be influenced by social desirability bias, leading them to provide responses that they believe are more socially acceptable rather than their true feelings. As a result, there is a need for a more comprehensive approach to measuring patient motivation during VR rehabilitation.

Our study proposes the use of camera-based sensors as an alternative solution for measuring patient engagement during VR exercises. By capturing and analyzing facial expressions, we aim to use emotion recognition technology to provide insight into a participant's emotional state and level of engagement. Unlike other sensors such as EEG and GSR[43, 45–47] which may be more expensive, camera-based sensors can capture images of the participant's face without the need for specialized equipment. This approach offers a non-invasive and relatively low-cost method for measuring patient engagement in VR exercises. Using neural networks to analyze the data collected from these sensors, our study aims to demonstrate the effectiveness of this approach for measuring patient engagement in VR serious games. Overall, this approach has the potential to inform the development of more engaging and effective

VR exercises that can better support patient motivation and treatment adherence. Our study aims to explore the feasibility and effectiveness of using emotion recognition technology to measure motivation in serious games. By employing our proposed approach, we hope to demonstrate the advantages of using neural network-based emotion recognition for measuring motivation in serious games. Overall, this approach has the potential to improve our understanding of how motivation operates in serious games during functional rehabilitation.

4 System Design

In this study, we have proposed an innovative architecture specifically designed for evaluating motivation in post-stroke patients during upper limb rehabilitation using serious games. This proposal consists of two main modules, as depicted in figure 1.

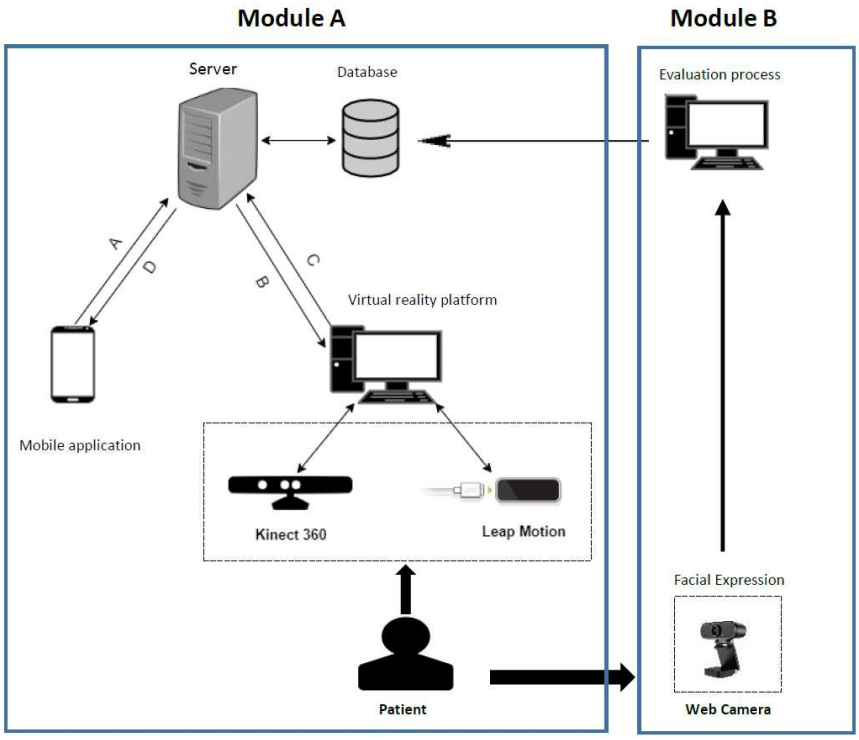

Fig. 1 Architectural Design for Post-Stroke Upper Limb Rehabilitation

4.1 Module (A): Serious game process

This module consists of four key steps: account initialization, serious game personalization, data storage, and Monitoring Patient Progress.

4.1.1. Account initialization (Step A)

The doctor creates a patient profile, which includes medical information about the patient, such as the hemiplegic side, the date of the stroke and a summary of the workup as shown in figure 2. This information is then transmitted to a server and stored in the database.

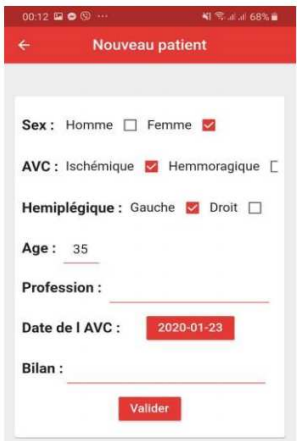

Fig. 2 Mobile interface

4.1.2. Serious game personalization (Step B)

At the beginning of each rehabilitation session, the patients identify themselves using an ID generated during profile creation. The platform then receives this information, allowing for a personalized rehabilitation experience. For example, the patient's gender is reflected in the avatar representing him/her, and if she/he is hemiplegic on the right side, the virtual scene is adjusted accordingly. The rehabilitation therapist can also personalize the session by selecting the difficulty level and determining the duration. During the rehabilitation session, a Microsoft Kinect camera is used as a depth image sensor to capture the patient's movements. Virtual hand movements are performed using data received from the Leap Motion controller API.

4.1.3. Data Storage (Step C)

At the end of each rehabilitation session, the VR platform communicates with the Backend to store important information about the rehabilitation session, including the score and duration of the session. This information is essential for monitoring the patient's progress and other details, such as the moment when the patient became fatigued, are also recorded.

4.1.4. Monitoring Patient Progress (Step D)

A Physiatrist communicates with the BackEnd to monitor the progress of patients, represented by a graph that shows the score over the course of rehabilitation.

4.2. Module (B): Evaluation process

During the VR-based rehabilitation sessions, the camera integrated into the platform (figure 1 module B) captures the facial expressions of the patients, enabling the evaluation of their level of motivation and enjoyment. Specifically, our focus lies in the recognition of facial emotions exhibited by the patients, utilizing convolutional neural networks (CNN) and the DeepFace model [48]. By employing these advanced technologies, we aim to accurately assess the emotional responses of patients and gain valuable insights into their motivation and engagement during the rehabilitation process. Recognizing facial emotions of patients using convolutional neural networks (CNNs) is a technique used in computer vision to detect and classify facial expressions. CNNs are a type of deep learning algorithm that can analyze visual input data, such as images or videos, to automatically extract features and learn patterns that can be used to recognize specific emotions. The process of recognizing facial emotions using CNNs in our approach typically involves the following steps:

4.2.1. Capturing and interpretation

Collecting images or videos containing facial expressions of patients. These images are obtained from camera-sensors.

4.2.2. Emotion recognition

A.Preprocessing: Before feeding the images to the CNN, they are preprocessed by resizing, cropping, and normalizing them to a fixed size (1616) to reduce computational complexity using Mood me framework [49] as showing in (figure 3).
B.Recognition: After data preprocessing, we employed the method proposed in [50] to classify the patient's facial expressions into various emotion categories, including positive, negative, neutral (figure 4). Our previous work [7] involved the use of the combination of MoodMe Emotions Barracuda SDK [49] for pre-processing, and DeepFace[48], for emotion classification, DeepFace a neural network model developed by Facebook's AI Research team in 2014. The recognition module (MoodMe + DeepFace) has demonstrated its capabilities in our previous research [7]. In this paper, our emphasis is on using emotion recognition to estimate the level of motivation in patients during serious games. For more details on our approach and its effectiveness, refer to our paper[7].

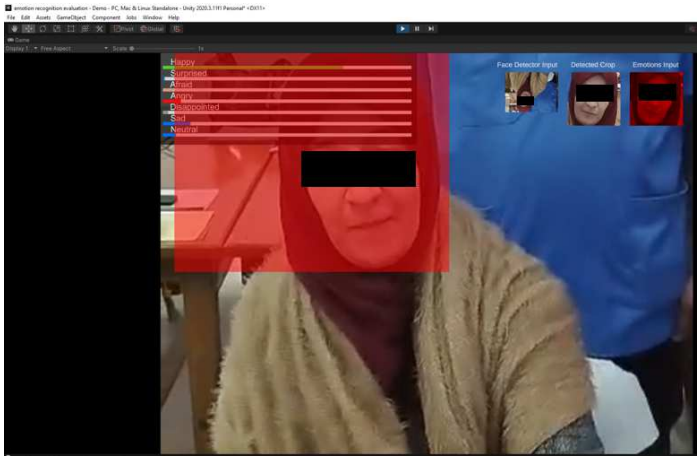

Fig. 3 Patient Facial Recognition Screenshot Using Moodme Framework

During the training phase, the user's facial expressions are captured by a webcam and analyzed to calculate the probabilities of being negative (Per(negative)), positive (P(positive)), and normal (P(normal)) for each frame. These probabilities are computed using equations (1), (2), and (3):

$$P(\text{positive}) = P(\text{happy}) + \frac{P(\text{surprised})}{2} \tag{1}$$

$$P(\text{negative}) = P(\text{sad}) + P(\text{angry}) + P(\text{disgust}) + \frac{P(\text{surprised})}{2} + P(\text{Afraid}) \tag{2}$$

$$P(\text{normal}) = P(\text{neutral}) \tag{3}$$

The motivation evaluation is done every Time = T, we must therefore calculate the average emotion (positive, negative, neutral) at each T. (see Equations (1)).

$$\text{Average} = \frac{\sum P_i(\text{Emotion})}{n} \tag{4}$$

Average = $\{APos, ANeg, ANeu\}$, Emotion = $\{positive, negative, normal\}$,
N: Number of frames in T.
Thus, the result of process will be, Max(APos, ANeg,ANeu).

In this study, we will employ classification techniques to assess the level of patient motivation with the serious game (figure 4). Based on the classification results, the following interpretations can be made:

- Positive emotion: The patient is motivated.
- Negative emotion: The patient is unmotivated.

- Neutral emotion: The patient's motivation level is neutral.

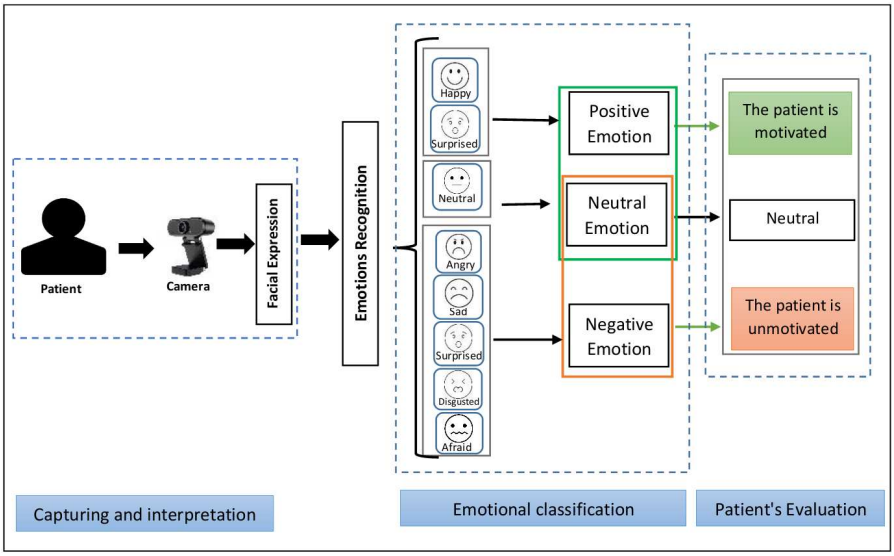

Fig. 4 Architecture for Assessing Patient Motivation in Serious Games through facial Emotion Recognition

5 Virtual environmental scenes

In this study, we focused on three key aspects in the design of five serious games for upper limb rehabilitation after a stroke [51]. Firstly, we developed a system that incorporates real rehabilitation exercises, ensuring that the games mimic the movements and actions performed during conventional rehabilitation sessions. Next, we integrated game design principles into the development process, incorporating elements such as goal setting, feedback, and rewards to enhance engagement and motivation. Lastly, we created an immersive environment that simulates familiar surroundings, providing a sense of comfort and familiarity to the patients.
In our virtual rehabilitation platform, we used two devices to enhance the user experience and maximize the therapeutic benefits for patients. The first device is the leap motion, which enable natural hand tracking, offering precise and responsive control of patient's movements. The second device is the kinect, which uses motion-tracking technology to track the patient's full body movements, providing a more immersive experience.

5.1 Serious games based on leap motion

5.1.1 Bowling

This serious game is based on a functional rehabilitation exercise involving the use of therapeutic putty to exercise the hand (figure 5.b). In the game, the goal is to

try to knock down all six bowling pins (strike) within ten seconds (figure 5.a); this is done in two phases, the first phase focuses on maximizing the force gauge (figure 5.a) by opening and closing the hand (gripping), and the second phase focusing on orienting the throwing of the bowling ball through gestures of flexion/extension on the sagittal plane (figure 5.c). The yellow arrow helps the patient to aim and provides visual feedback.

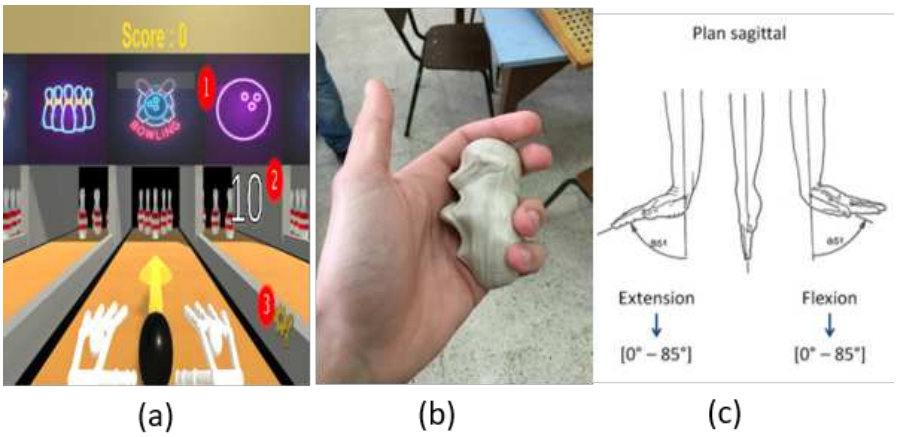

Fig. 5 (a). The virtual environment of the serious game simulating the exercise Bowling . (b). Exercise with therapy putty. (c). Flexion / Extension of the wrist.

5.1.2 Fruit collection

This game is mainly based on two exercises. It targets various joints, including the hand, elbow and shoulder. In the first exercise, the patient uses a therapeutic paste (figure 5-b), and in the second the exercise consists of flexing/extending the shoulder in the sagittal plane, as shown in figure 6-b. In the exercise illustrated in figure 6-a, the patient is asked to raise his arm and pick up as much fruit as possible. The score increases each time the fruit falls into the basket. To obtain the highest score, the patient must repeat this gesture several times.

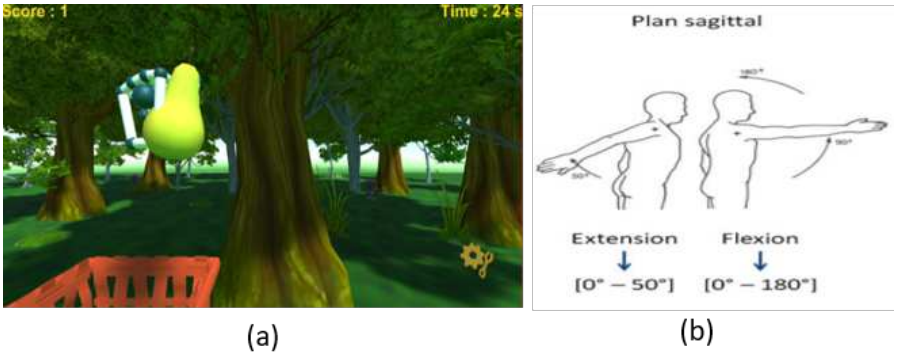

Fig. 6 (a). Serious game interface Fruit collection. (b). Flexion / Extension of the shoulder.

5.1.3 Arcade

The rehabilitation game is based on an exercise used in occupational therapy. The exercise involves simply taking a bottle of water and pouring its contents into cups. The game focuses on the pronation and supination movement of the forearm, as demonstrated in figure 7.b. The patient must reach for the glass to grasp it, then turn the wrist to pour the ball into the cup whose color corresponds to that of the ball. If the ball lands in the right cup, the score is incremented and the scene is reset with new balls. The game features two levels of difficulty: the first level consists of a single ball, while the second level includes two balls, as illustrated in figure 7.a.

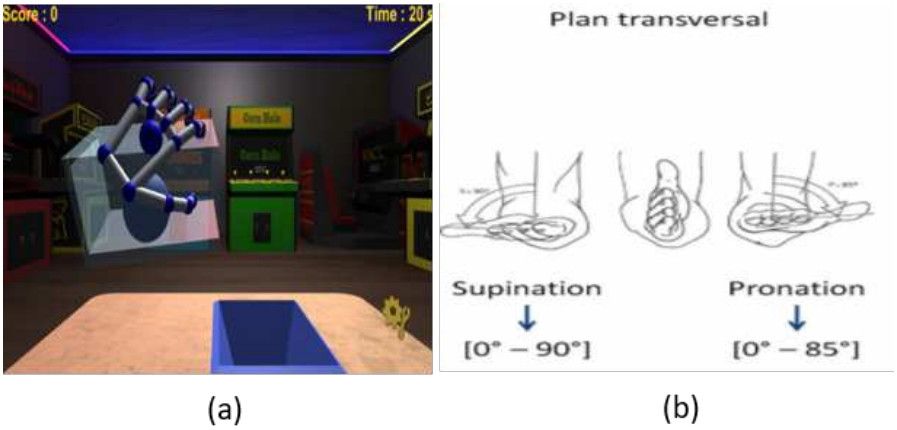

Fig. 7 (a). Serious game interface Arcade .(b). Pronation / Supination of the forearm.

5.2 Serious games based on Kinect 360

5.2.1 Hammer

In this game, the patient sits in front of a table and picks up an object (a water bottle, for example) using his hemiplegic hand. He then slides the object across the table, keeping it as far away from him as possible. The exercise is repeated by returning the object to its starting position and repeating the process in the opposite direction (figure 8.b).The patient stands in front of a Kinect camera, which captures and imitates his or her movements through an avatar. The aim of the game is to use a hammer to hit a mole (represented by a red sphere) that appears in one of several holes (figure 8.a). The patient is motivated by a reward system that offers a 5-point bonus if he or she manages to complete the exercise within 6 seconds of the mole's appearance.

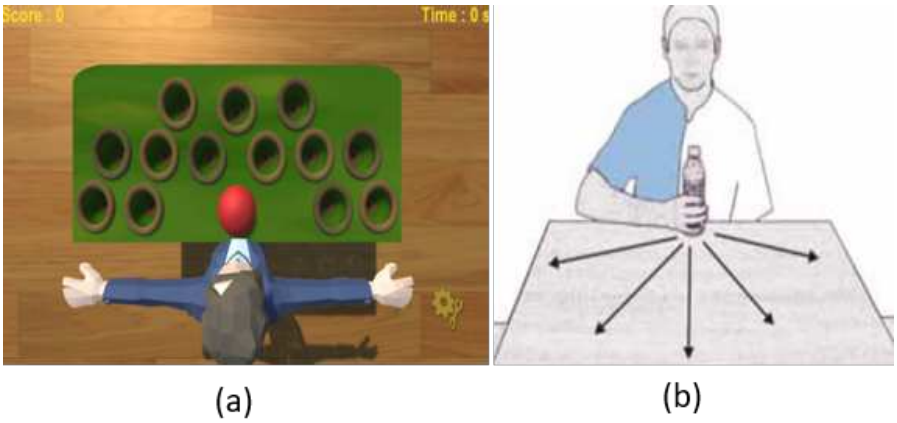

Fig. 8 (a). Serious game interface Hammer.(b). Rehabilitation exercise.

5.2.2 Attic

The serious game is designed to simulate an attic environment in which the patient must put away objects from shelves and place them in a cardboard box in front of him. The patient stands in front of a Kinect camera and uses their movements to control the arms of an avatar, as depicted in figure 9.a. The game features two levels of difficulty, each aimed at stimulating different aspects of shoulder movement. In Level 1, all objects are at the same height, which primarily stimulates internal and external rotational movement of the shoulder in the transverse plane, as illustrated in figure 9.b. In Level 2, some objects are placed at a higher height, requiring the patient to abduct and adduct their shoulder in the frontal plane, as depicted in figure 9.c.

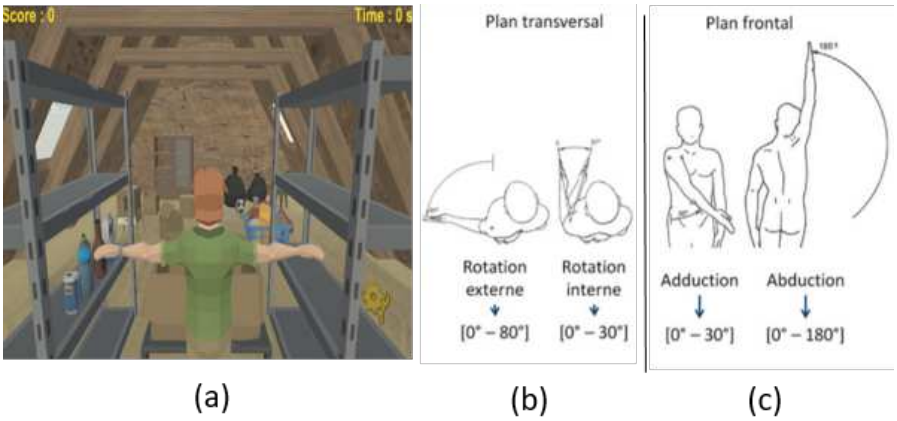

Fig. 9 (a). Serious game interface Attic . (b). Internal and external rotation of the shoulder. (c). Adduction and abduction of the shoulder.

6 Results and discussion

To analyze whether VR is motivating patients or not, we need to use the data given (table 2), which consists of facial expressions of patients while using VR in rehabilitation exercises. The data has been categorized into three groups - unmotivated, motivated, and neutral, based on the facial expressions observed. Based on the results provided, it appears that the majority of patients in the study were motivated during their VR rehabilitation exercises, with an average motivated percentage of around 59.2% . However, there is variability in the levels of motivation across patients, with some patients being more motivated than others. For example, patient 6 had a motivated percentage of 74%, while patient 15 had a motivated percentage of only 38% (figure 10).

It is important to acknowledge that the accuracy and reliability of emotion recognition technology can be influenced by several factors, such as lighting conditions, position and angle of the camera during photography, and the quality of the camera-based sensors used. Additionally, the level of pain experienced by the patient during exercise can also affect the accuracy of the results. Therefore, we incorporated additional factors to obtain more accurate results. Specifically, we considered the type of hypothetical exercise and its level of difficulty, and we calculated the average outcomes for each patient. Subsequently, we compared these average results with the best possible outcomes achievable in each game. We also took into account the number of VR sessions carried out by the patients (as shown in table 3).Based on the data provided in table 3, it appears that the patients generally scored lower than the reference scores, which is expected due to the side effects of the stroke such as difficulties and motor problems. In addition, most of the results were low in the first rehabilitation sessions, which could be attributed to the initial impact of the stroke. However, it's important to note that the average scores were not significantly lower,

Table 2 Facial Expressions Analysis of Post-Stroke Patients during VR Rehabilitation Exercises.

Patients	Average Percentage of Unmotivated (%)	Average Percentage of motivated (%)	Average Percentage of Neutral (%)
P1	34,65055661	65,34944474	45,04358075
P2	53,85360277	46,1463974	65,36485971
P3	35,44309805	64,55690093	62,55397628
P4	40,71084457	59,28915474	75,58552166
P5	45,0820438	54,91795608	66,65752396
P6	25,96601136	74,03398733	28,73164733
P7	30,63608932	69,36390918	45,66000128
P8	17,66681636	82,33318164	32,42917159
P9	52,32843587	47,67156285	65,10375442
P10	51,2236632	48,77633482	68,54950775
P11	52,83501459	47,16498606	74,62704784
P12	40,03407193	59,96592867	52,19965177
P13	37,99256933	62,00742926	54,55583544
P14	32,43649433	67,56350502	75,36359133
P15	61,69772172	38,30228005	41,03966182
P16	35,52996712	64,47003171	42,89605272
P17	40,57703559	59,42296256	65,81930949
P18	32,0814763	67,91852271	32,76770587
P19	36,83591933	63,16408016	44,46429237
P20	58,63259171	41,36740832	85,96249547

and there were instances where patients achieved scores that were relatively close to the reference scores. For instance, patients 6 and 8 showed higher levels of motivation during the game, with motivation rates estimated at 74% and 82%, respectively. This explains their high scores, which were closer to the reference scores. Where the patient 6 had an average score of 87 out of 107 at the first level (figure 11) and 69 out of 86 at level 2, while patient 8 had an average score of 40 out of 43 on the first level and 31 out of 36 at level 2 (figure 13). It's also worth mentioning that the level of the game played by the patients could have affected their performance, as higher levels typically require more advanced skills and coordination.

Additionally, the motivational state of the patients seems to have had an impact on their scores, with motivated patients generally performing better than unmotivated ones (figure 12 and 14). With the exception of patients 11 and 15, as shown in table 2 and figure10, we observed that their estimated unmotivation rates were 52 % and 61%, respectively. However, they still achieved high scores, likely due to the pain they experienced in the elbow (patient 15) and wrist (patient 11). Notably, these patients exhibited non-motivating facial expressions during pain, as indicated in Equation 2. Despite this, their scores were comparable to the reference scores, demonstrating the effectiveness of VR to motivate the patient to do and finish exercises. The results also showed that out of 20 patients, we found two patients who had obtained low scores in the rehabilitation games. Patient 2 obtained an average of 45 out of 140 at the first level and 28 out of 63 at level 2, while patient 20 obtained a score of 71 out of 140 at the first level and 22 out of 63 at level 2. The two patients were not motivated, with

Table 3 Comparison of Patient Scores with Reference Scores in VR Rehabilitation Exercises. (G1: Bowling, G2: Fruit collection, G3: Arcade, G4: Hammer, G5: Attic. Male: M, Female: F)

Patients	Games	sexe	Number of VR-rehab sessions	Level	Patient average score	Reference average score	Level	patient average score	Reference average score
P1	G2	M	16/16	1	106	140	2	43	63
P2	G2	F	9/16	1	45	140	2	28	63
P3	G5	M	16/16	1	36	50	2	33	42
P4	G4	F	15/16	1	26	43	2	25	36
P5	G3	M	16/16	1	13	22	2	8	17
P6	G1	M	16/16	1	87	107	2	69	86
P7	G2	M	16/16	1	108	140	2	51	63
P8	G4	M	16/16	1	40	43	2	31	36
P9	G5	F	13/16	1	21	50	2	19	42
P10	G5	M	15/16	1	31	50	2	28	42
P11	G1	M	16/16	1	82	107	2	72	86
P12	G5	M	16/16	1	33	50	2	31	42
P13	G4	M	16/16	1	34	43	2	22	36
P14	G5	M	16/16	1	37	50	2	31	42
P15	G3	M	16/16	1	17	22	2	11	17
P16	G3	F	15/16	1	13	22	2	7	17
P17	G1	M	16/16	1	88	107	2	61	86
P18	G2	F	16/16	1	111	140	2	49	63
P19	G1	F	16/16	1	79	107	2	56	86
P20	G2	F	10/16	1	71	140	2	22	63

estimated unmotivated rates of 53% for patient 2 and 58% for patient 20. It is also important to note the number of VR sessions completed by each patient. As indicated in table 3, the majority of patients successfully finished the 16 VR exercise sessions, demonstrating their commitment to the rehabilitation process. This adherence to the recommended sessions reflects the motivating effect of VR on post-stroke patients, encouraging their active participation in the prescribed exercises. It is worth mentioning that patients 2 and 20 did not complete the full set of sessions. While their lack of completion indicates a lack of motivation towards functional rehabilitation (figure 10), it is essential to consider individual circumstances and potential reasons behind their incomplete participation and non-motivation. Further investigation and assessment are required to understand the specific factors that may have influenced their inability to finish the sessions. Nonetheless, the overall trend observed in the majority of patients completing the recommended sessions highlights the potential of VR as a powerful tool for motivating post-stroke patients to engage in rehabilitation exercises. By providing an immersive and interactive environment, VR offers a unique approach that can captivate and inspire patients to actively participate in their recovery journey, guided by the recommendations of healthcare professionals.

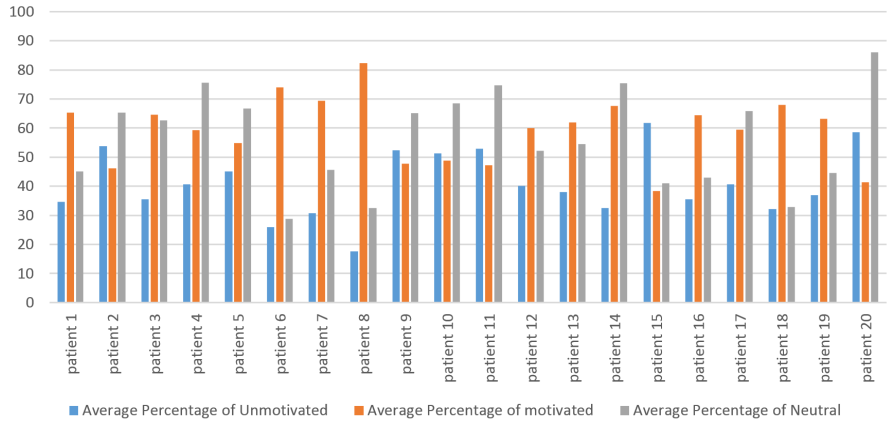

Fig. 10 Emotional Response of Post-Stroke Patients during VR Rehabilitation Exercises.

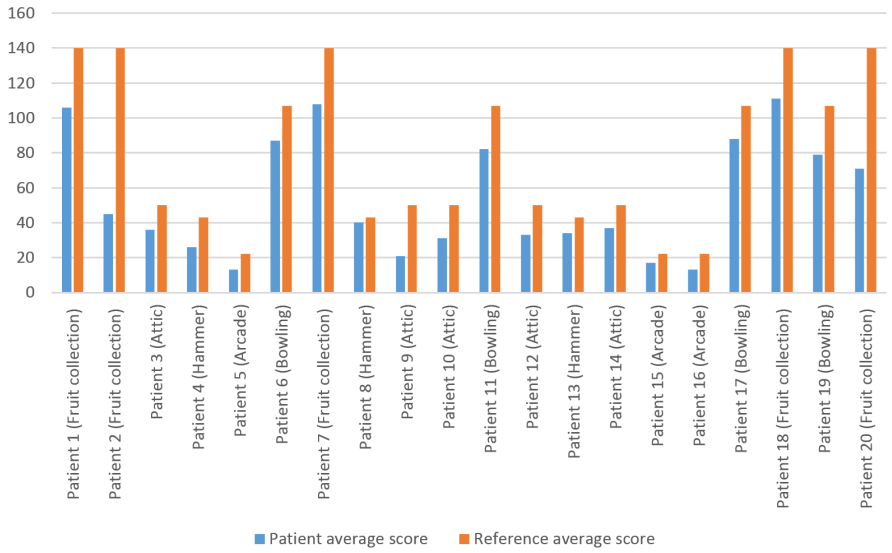

Fig. 11 Average Scores of Patients at Level 1 in VR Rehabilitation Exercises.

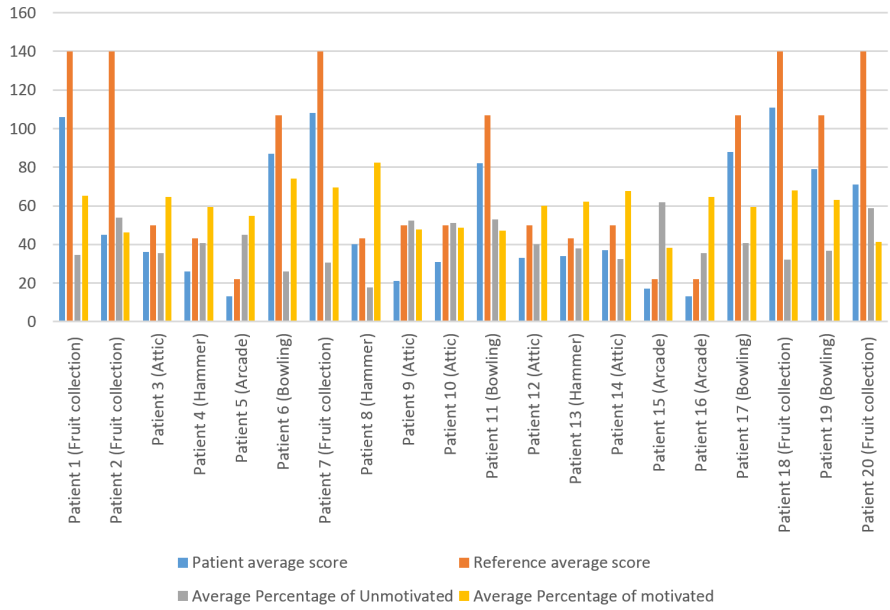

Fig. 12 Performance of Patients at Level 1 based on Motivational State in VR Rehabilitation Exercises.

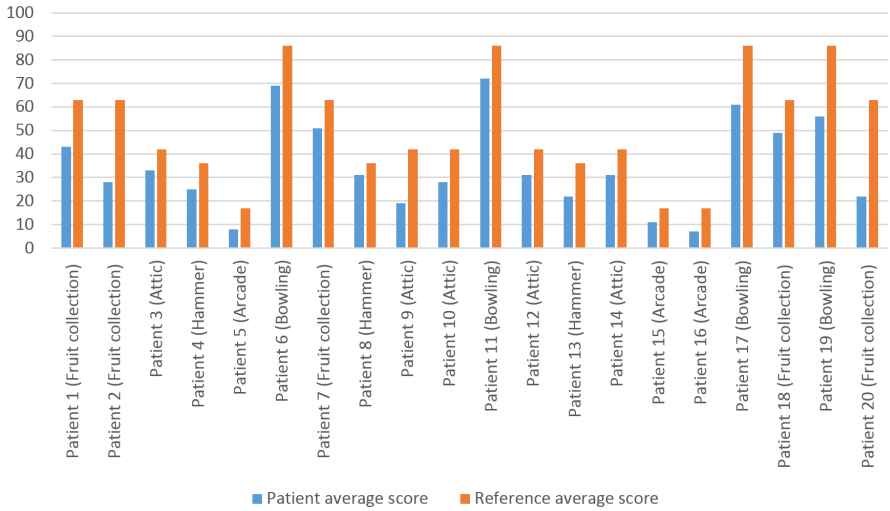

Fig. 13 Average Scores of Patients at Level 2 in VR Rehabilitation Exercises.

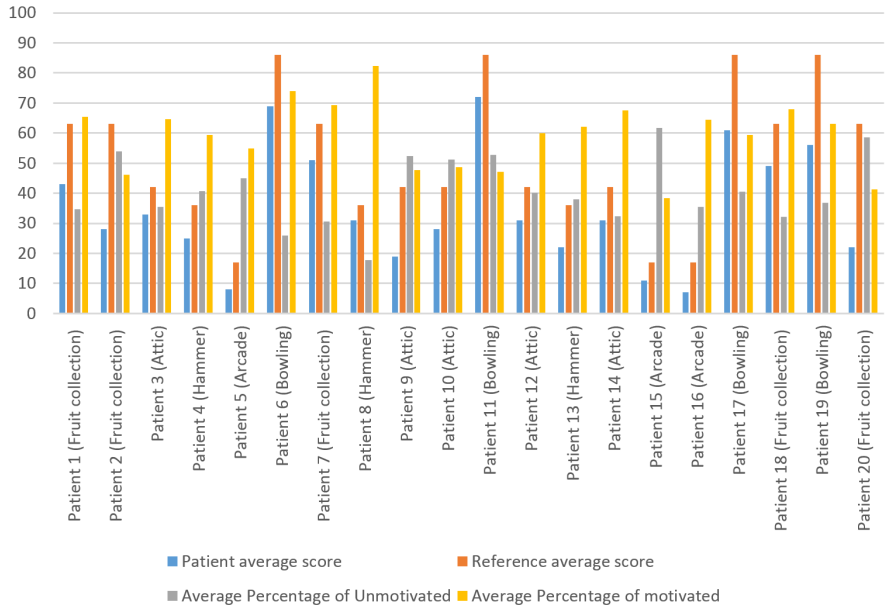

Fig. 14 Performance of Patients at Level 2 based on Motivational State in VR Rehabilitation Exercises.

7 Conlusion

This article presents a comprehensive study on the effectiveness of VR in post-stroke rehabilitation, with a specific focus on patient motivation and engagement. The inconclusive results found in previous studies regarding the effectiveness of VR can be attributed to the complex nature of motivation as a multifaceted construct, requiring the development of appropriate measurement tools and protocols for accurate assessment. This study contributes to addressing this challenge by proposing a novel evaluation approach that incorporates camera-based sensors and emotion recognition technology to gain insights into patient engagement during VR exercises. Unlike previous studies, this research recognizes the significance of a homogeneous sample of patients. By recruiting a diverse sample of 20 patients, both male and female, who had experienced left hemiplegia as a result of an ischemic stroke, the study aims to provide more reliable and applicable findings. The findings indicate that VR rehabilitation exercises are effective in motivating a significant number of stroke patients. Furthermore, the commitment of the majority of patients to complete the recommended sessions further demonstrates the positive influence of VR on patients, encouraging their active participation in the exercises. This commitment reflects the motivating effect of VR and reinforces its potential as an effective intervention in post-stroke rehabilitation.

A PERSPECTIVE on this work is that the successful implementation of VR in post-stroke rehabilitation, as demonstrated by the positive findings and patient engagement, raises intriguing possibilities for utilizing emerging technologies like the metaverse

world to enhance the immersive and interactive aspects of rehabilitation programs, potentially leading to even greater therapeutic outcomes for stroke patients.

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